Xinran ZHAO

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Education

- **City University of Hong Kong** *Aug. 2020 - Present*
  Ph.D. in Electrical Engineering Hong Kong SAR
  - Supervisor: Prof. Lin DAI
  - GPA: 4.15/4.30

- **Huazhong University of Science and Technology** *Sep. 2016 - Jun. 2020*
  Bachelor in Electronics and Information Engineering Wuhan, China
  - GPA: 3.86/4.0
  - Thesis: "Random access toward massive machine-to-machine communications: Performance analysis and protocol design", supervised by Dr. Yayu GAO.

Research Interests

- **Decentralized multiple access**: Modeling, analysis and optimal design.
- **Next-generation wireless communication networks**: Massive Internet of Things, low-latency communications, and distributed learning-based access design.

Academic Experience

- **Learning-Based Random Access: Design and Optimization [S1]** *Jul. 2024 - Present*
  - A simple Multi-Armed-Bandit (MAB) learning framework was presented for performance optimization of random access, based on which two random access schemes were proposed for throughput maximization, that is, MTOA-L with local rewards and MTOA-G with global rewards.
  - The proposed MTOA-L and MTOA-G were shown to both achieve the maximum throughput of 1, though with different short-term fairness performance.
  - It was demonstrated that by leveraging the queueing-theoretical analysis, the tradeoff between throughput and short-term fairness of MTOA-L and MTOA-G can be characterized and optimized by tuning the key parameters.

- **Random Access: Unified Modeling, Optimization and Comparison [J2]** *Oct. 2022 - Jun. 2024*
  - A unified analytical framework for random access was established, which incorporates various access design features including sensing-free or sensing-based, connection-free or connection-based, and backoff.
  - The unified framework enables not only the delay characterization and optimization for a given random access scheme, but also a comprehensive comparison of optimal delay performance across different schemes.
  - Useful criteria, such as the upper-bound of sensing time for benenficial sensing, were obtained and further applied to cellular networks.

- **Connection-Based Aloha: Modeling and Performance Optimization [J1]** *Aug. 2020 - Sep. 2022*
  - A scalable model was developed for connection-based Aloha.
  - Both the throughput performance and queueing delay performance were characterized, optimized, and compared with those for connection-free Aloha.
  - Conditions for beneficial connection establishment were discussed and further applied to cellular networks.

TEACHING EXPERIENCE

- **Tutor for Final Year Projects**
  - Optimal Backoff Tuning for Massive Access of Machine-type Devices in 5G Networks *Fall 2024*
  - Massive Access for Beyond-5G Communications Networks: Connection-based or Connection-free? *Fall 2021, Fall 2022, Spring 2024*
  - Energy-Efficient Massive Access for Machine-to-Machine Communications *Fall 2020*
- **Teaching Assistant**
  - Data Engineering and Learning Systems *Spring 2024*
  - Mobile Communication and Networks *Spring 2022, Fall 2022, Fall 2023*
  - Principles of Communications *Fall 2021, Spring 2023*

PROFESSIONAL EXPERIENCE

- **Intelligent Control System of Tobacco Baking Room** *Mar. 2019 - Jan. 2020*
  Team leader of a collaboration project between Dian Group (student tech team in HUST) and China Tobacco Wuhan, China
  - **Back-End Development**: Developed the server-side APIs to implement the functions of data reception from IoT terminals, status monitoring of baking rooms, historical data statistics, remote control, etc.
  - **System Debugging and Deployment**: Conducted the joint debugging between the server and IoT embedded terminals, the communications between which were built upon the LoRaWAN architecture. Deployed the system in practical tobacco-baking scenarios.
  - **Skills**: Python Django, MySQL, Redis and LoRaWAN.
- **Online Exam System** *Sep. 2018 - Jan. 2020*
  Member of a project at Dian Group Wuhan, China
  - **Back-End Development**: Developed the server-side APIs to realize the functions of question bank management, examinee information management, online examination, grading, etc.
  - **Skills**: Python Flask and MySQL.

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, S=IN SUBMISSION

[S1] Xinran Zhao and Lin Dai, "Throughput-optimal random access: A queueing-theoretical analysis for learning-based access design," submitted for publication. (Preprint: https://arxiv.org/abs/2504.03178)

[J2] Xinran Zhao and Lin Dai, "To sense or not to sense: A delay perspective," to appear in *IEEE Transactions on Communications*.

[J1] Xinran Zhao and Lin Dai, "Connection-based Aloha: Modeling, optimization, and effects of connection establishment," *IEEE Transactions on Wireless Communications*, vol. 23, no. 2, pp. 1008–1023, Feb. 2024.

[C1] Xinran Zhao and Lin Dai, "To sense or not to sense: A delay perspective," IEEE International Conference on Communications, Denver, CO, USA, 2024, pp. 2318-2323.

AWARDS

- Research Tuition Scholarship *City University of Hong Kong, Sep. 2023*
- EE Graduate Research Seminar Award - 3nd Prize *Dept. of EE, City University of Hong Kong, May 2022*
- Outstanding Undergraduate Thesis Award *Huazhong University of Science and Technology, Jun. 2020*

SKILLS

- **Mathematical Tools:** Probability theory, queueing theory, stochastic process, reinforcement learning.
- **Programming Languages:** Python, MATLAB, C.
- **Web Technologies:** Python Flask, Python Django, MySQL, Redis.

ADDITIONAL INFORMATION

- **Languages:** Mandarin (native) and English (IELTS 7.0)